

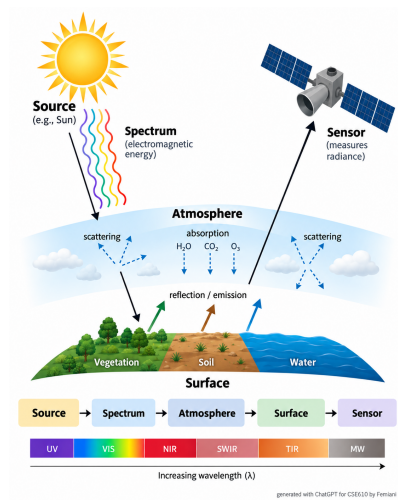
IRS Chapter 2: Electromagnetic Radiation

Dr. John C. Femiani

CSE620B: Computer Vision for Remote Sensing
Miami University

Why This Chapter Matters

- ▶ Most *conventional* Earth observation is an **EM measurement problem**, not just an imaging problem.
- ▶ Sensor design depends on **where energy exists, how it travels, and how surfaces modify it**.
- ▶ This chapter builds the chain: **source** → **spectrum** → **atmosphere** → **surface** → **sensor**.



Guiding Question

Guiding Question

Which wavelengths carry usable information about Earth, and what distorts that information before it reaches the sensor?

Roadmap Checkpoint

Foundations of EM Radiation

Spectrum Regions and Measurement

Radiation Laws and Thermal Sensing

Atmospheric Interactions

Surface Interactions and Sensor Signal

(Electromagnetic) Waves, Phase, and Quadrature

► Sinusoidal wave

$$A(t) = A_0 \cos(\omega t + \phi)$$

$$\underbrace{A_0}_{\text{amplitude}} \quad \underbrace{\omega}_{\text{angular frequency}} \quad \underbrace{\phi}_{\text{phase offset}}$$

► Quadrature = quarter-cycle ($90^\circ = \pi/2$) shift

$$\cos \theta \longleftrightarrow \sin \theta$$

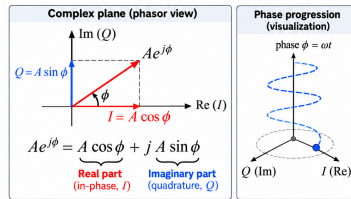
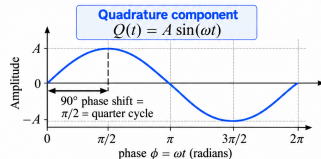
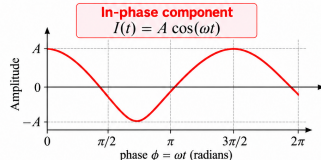
► I/Q coordinates

$$I = A_0 \cos \phi, \quad Q = A_0 \sin \phi$$

► Complex shorthand

$$A_0 e^{j\phi} = I + jQ$$

► Re/Im are mathematical coordinates—not separate physical fields.



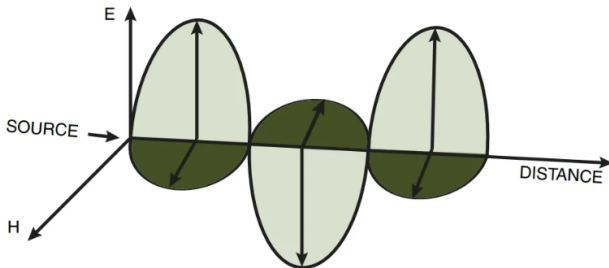
generated with ChatGPT for CSE610 by Farnian

Electromagnetic Radiation as a Sensing Medium

- ▶ EM radiation consists of coupled **electric** and **magnetic** fields that propagate through space. This is distinct from the real and imaginary components of a complex wave representation.
- ▶ For a plane wave,

$$\underbrace{\mathbf{E}}_{\text{electric field}} \perp \underbrace{\mathbf{H}}_{\text{magnetic field}} \perp \underbrace{\mathbf{k}}_{\text{direction}}, \quad \frac{|\mathbf{E}|}{|\mathbf{H}|} = \underbrace{\eta}_{\text{intrinsic impedance of the medium}}$$

- ▶ Its key sensing properties are **wavelength**, **frequency**, **amplitude**, and **polarization**.

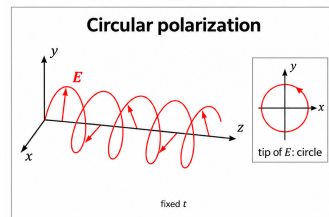
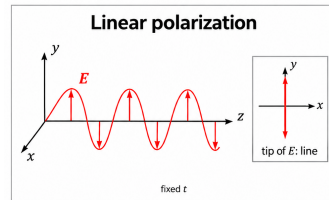


Polarization

- Polarization = behavior of the transverse **electric-field vector $\mathbf{E}$**
- **Linear:** $\mathbf{E}$ oscillates along one fixed direction
- **Circular:** two orthogonal E components, equal amplitude, 90° phase offset

$$E_x = A \cos(\omega t - kz), \quad E_y = A \sin(\omega t - kz)$$

- At fixed position: tip of $\mathbf{E}$ traces a **circle**
- At fixed time: $\mathbf{E}$ traces a **spiral through space**

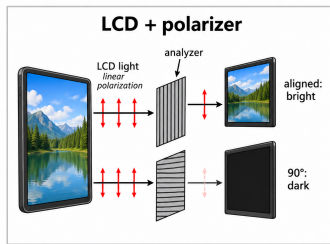
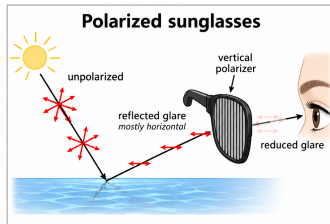


z = propagation direction (space), not time

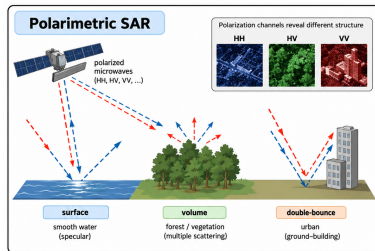
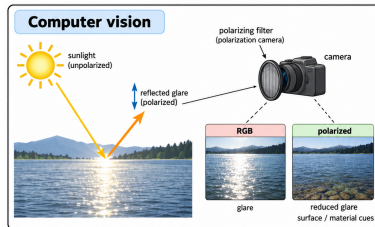
generated with ChatGPT for CSE610 by Femiani

Why Polarization Matters for Sensing

Light tends to change polarization when it reflects, refracts, or scatters.



generated with ChatGPT for CSE610 by Femiani



generated with ChatGPT for CSE610 by Femiani

Where the Energy Comes From

Natural sources

- ▶ Solar radiation drives most reflective imaging.
- ▶ Thermal motion drives longwave emission.
- ▶ Atomic and nuclear processes produce higher-energy bands.

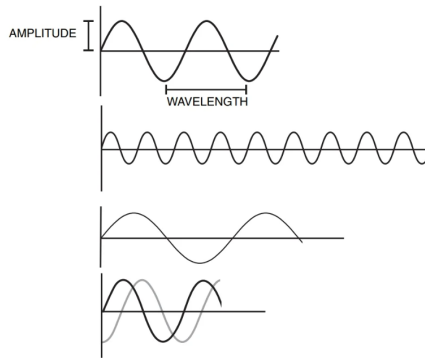
Active sources

- ▶ Radar systems illuminate the scene directly.
- ▶ Useful when **sunlight is absent** or **cloud-penetrating bands** are needed.
- ▶ The instrument controls the wavelength and timing.

Wave Quantities We Reuse All Semester

- ▶ **Wavelength** λ : crest-to-crest spacing.
- ▶ **Frequency** ν : cycles per second.
- ▶ **Amplitude**: energy magnitude.
- ▶ **Phase**: alignment between waves.

$$c = \lambda \nu$$



Fast Example: Same Speed, Different Information

- ▶ All EM waves travel at speed c in vacuum.
- ▶ Shorter λ means higher ν and usually higher photon energy.
- ▶ Visible, thermal IR, and microwave sensing differ mostly by **where they sit on this tradeoff**.
- ▶ That is why band choice changes penetration, scattering, heating, and material contrast.

Roadmap Checkpoint

Foundations of EM Radiation

Spectrum Regions and Measurement

Radiation Laws and Thermal Sensing

Atmospheric Interactions

Surface Interactions and Sensor Signal

Measurement Units Used in Remote Sensing

Length units

Unit	Scale
meter (m)	10^0
millimeter (mm)	10^{-3}
micrometer (μm)	10^{-6}
nanometer (nm)	10^{-9}

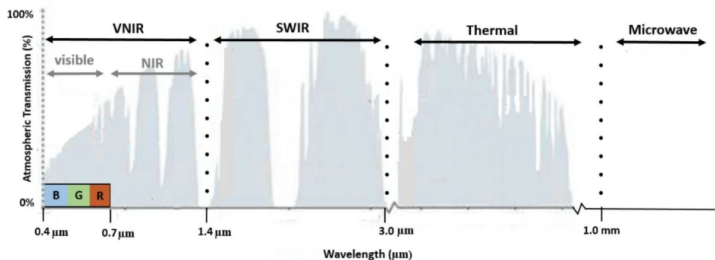
Frequency units

Unit	Scale
Hz	10^0
MHz	10^6
GHz	10^9
THz	10^{12}

Remote sensing literature mixes wavelength and frequency conventions; both appear in sensor specifications.

The Spectrum Is Continuous; the Labels Are Human

- ▶ Names like **visible**, **NIR**, **SWIR**, and **microwave** are convenient bins.
- ▶ The physics changes continuously; the category boundaries are conventional.
- ▶ For remote sensing, the practical question is **which intervals pass through the atmosphere and interact strongly with the target**.



Bands That Matter Most for Earth Observation

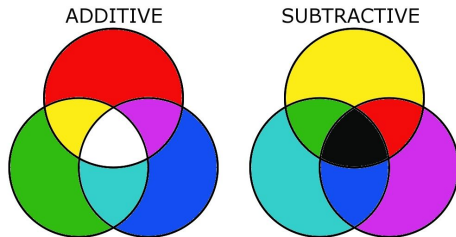
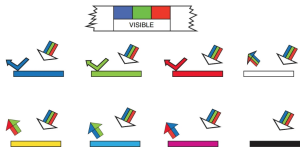
- ▶ **Visible:** about 0.4–0.7 μm .
- ▶ **NIR:** about 0.7–1.4 μm .
- ▶ **SWIR:** about 1.4–3 μm .
- ▶ **Thermal IR:** about 8–14 μm .
- ▶ **Microwave:** millimeters to meters.

Typical uses

- ▶ Visible: scene appearance, color, mapping.
- ▶ NIR: vegetation vigor.
- ▶ SWIR: moisture, minerals, burn scars.
- ▶ Thermal IR: surface temperature.
- ▶ Microwave: radar, all-weather imaging.

Visible Light Is a Tiny but Familiar Window

- ▶ Human vision spans only a narrow slice of the total spectrum.
- ▶ Blue, green, and red are useful reference bands, not the whole story.
- ▶ Remote sensing gains leverage by moving beyond human-visible wavelengths.



Roadmap Checkpoint

Foundations of EM Radiation

Spectrum Regions and Measurement

Radiation Laws and Thermal Sensing

Atmospheric Interactions

Surface Interactions and Sensor Signal

Why Radiation Laws Matter

- ▶ Chapter 2 is not just vocabulary; it explains **why some bands contain temperature information**.
- ▶ Light shows both **wave behavior** and **photon energy quantization**.
- ▶ Remote sensing needs both views: propagation as a wave, emission and detection as energy packets.

Core Relationships

Photon energy

$$Q = h\nu$$

Higher frequency means **higher energy per photon**.

Radiometric quantities

- ▶ Flux: total power delivered.
- ▶ Irradiance: power per unit area.
- ▶ Exitance: emitted power per unit area.

These quantities let us move from qualitative pictures of light to measurable sensor signals.

Blackbodies and Real Surfaces

- ▶ A **blackbody** absorbs all incident radiation.
- ▶ Its emission depends only on temperature.
- ▶ Real materials emit less efficiently.

$$\varepsilon = \frac{M}{M_b}$$

Interpretation

- ▶ $\varepsilon = 1$: ideal emitter.
- ▶ $\varepsilon < 1$: real surface.
- ▶ Emissivity helps distinguish soil, water, vegetation, and metal surfaces.

Two Laws Behind Thermal Remote Sensing

Stefan–Boltzmann law

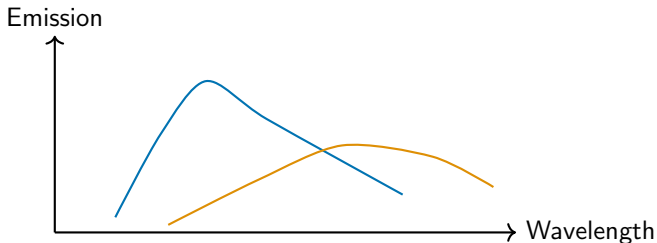
$$M = \sigma T^4$$

Total emitted energy rises sharply with temperature.

Wien's displacement law

$$\lambda_{\max} = \frac{b}{T}$$

Hotter objects peak at shorter wavelengths.



Worked Example: Why Earth Peaks in Thermal IR

- ▶ Earth-like temperatures are far below solar temperatures.
- ▶ Wien's law pushes the peak emission for Earth toward the **thermal infrared**.
- ▶ The Sun peaks near visible wavelengths, so reflective imaging mostly sees **solar illumination**.
- ▶ Thermal sensors instead estimate temperature from **Earth-emitted radiation**.

Roadmap Checkpoint

Foundations of EM Radiation

Spectrum Regions and Measurement

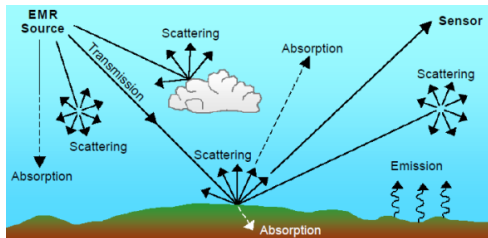
Radiation Laws and Thermal Sensing

Atmospheric Interactions

Surface Interactions and Sensor Signal

Atmosphere as a Filter, Not Empty Space

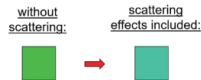
- ▶ Radiation to a satellite may travel through a long atmospheric path before detection.
- ▶ Along that path, energy can be **scattered**, **absorbed**, **transmitted**, or **refracted**.
- ▶ Atmospheric correction exists because the sensor sees **surface signal plus atmospheric contamination**.



Scattering Changes Apparent Brightness and Color

- ▶ **Rayleigh scattering:** small particles, strong at short wavelengths.
- ▶ **Mie scattering:** larger particles, haze, dust, smoke, cloud droplets.
- ▶ Scattering adds path radiance and reduces surface contrast.

1. Shift in COLOR:



2. Shift in BRIGHTNESS:



3. Loss of RESOLUTION:

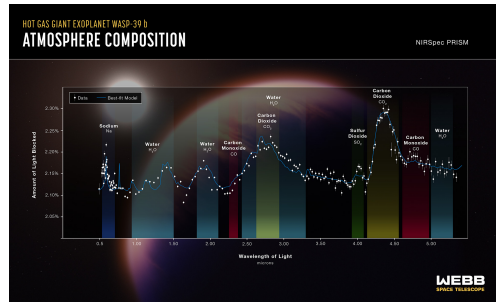


4. Loss of CONTRAST:



Absorption and Atmospheric Windows

- ▶ Ozone blocks much UV.
- ▶ Water vapor and CO_2 absorb major infrared intervals.
- ▶ Transmission windows are the bands remote sensors exploit.



Refraction and Practical Consequences

- ▶ Changing air density bends propagation paths.
- ▶ Refraction can shift apparent object position.
- ▶ Combined lesson: pick bands with good transmission, then model scattering and geometry corrections.

Design implication

Sensor bands are chosen around atmospheric windows because the atmosphere decides what information survives the trip.

Roadmap Checkpoint

Foundations of EM Radiation

Spectrum Regions and Measurement

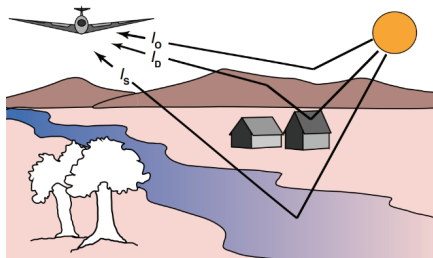
Radiation Laws and Thermal Sensing

Atmospheric Interactions

Surface Interactions and Sensor Signal

The Sensor Measures a Mixture

- ▶ The outgoing signal is not only reflected surface energy.
- ▶ It also includes atmospheric scattering and diffuse illumination.
- ▶ Interpretation depends on separating **illumination**, **material response**, and **path effects**.



Three Components of At-Sensor Radiance

Surface term

$$I_S$$

Reflected or emitted by the target.

Atmospheric terms

$$I_O, I_D$$

Path scattering and diffuse contributions.

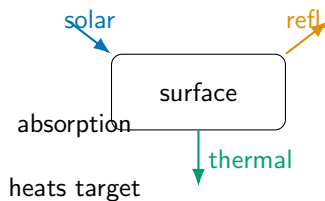
- ▶ Shadows often suppress direct reflection but keep diffuse illumination.
- ▶ That is why shaded pixels still contain class-dependent spectral information.

Reflection, Absorption, and Emission at the Surface

- ▶ Incoming solar energy is partitioned into **reflection**, **absorption**, and later **emission**.
- ▶ Reflective remote sensing emphasizes the shortwave part of this exchange.
- ▶ Thermal remote sensing emphasizes the emitted longwave response.
- ▶ Different materials partition energy differently, creating spectral signatures.

Energy Balance as the Link to Applications

- ▶ Vegetation, soil, water, and urban surfaces absorb and reradiate differently.
- ▶ That contrast supports land-cover mapping, water stress monitoring, burn assessment, and urban heat analysis.



Key Takeaways

1. EM radiation is described jointly by **wavelength**, **frequency**, and **energy**; remote sensing usually thinks in wavelength bands.
2. Useful sensing regions are determined by **radiation physics** and **atmospheric windows**, not by human vision.
3. The measured signal mixes **source energy**, **atmospheric effects**, and **surface reflectance or emission**; interpretation requires separating them.

References

-  Jensen, J. R. Introductory Digital Image Processing: A Remote Sensing Perspective.
-  Lillesand, T., Kiefer, R., and Chipman, J. Remote Sensing and Image Interpretation.
-  Planck, M. On the Theory of the Energy Distribution Law of the Normal Spectrum.
-  NASA Earth Observatory. Atmospheric windows, scattering, and thermal sensing reference materials.
-  NIST. Blackbody radiation and radiometric terminology reference pages.

Thank You

Questions?

We can now read later chapters as design choices about which part of this chain each sensor exploits.

Backup Slides

Backup: Unit Conversions Used Repeatedly

Length units

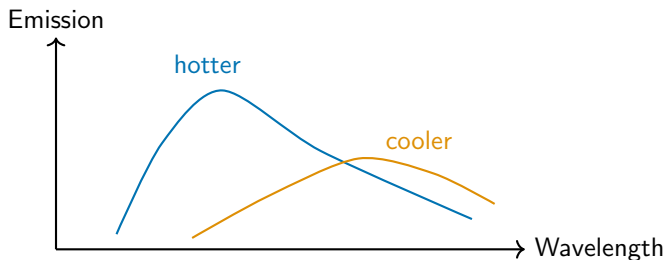
Unit	Equivalent
mm	10^{-3} m
μm	10^{-6} m
nm	10^{-9} m

Frequency units

Unit	Equivalent
kHz	10^3 Hz
MHz	10^6 Hz
GHz	10^9 Hz

Backup: Wien and Stefan–Boltzmann Together

- ▶ Hotter objects emit **more total energy**: $M = \sigma T^4$.
- ▶ Hotter objects also peak at **shorter wavelengths**: $\lambda_{\max} = b/T$.
- ▶ Sun-like temperatures peak near visible bands.
- ▶ Earth-like temperatures peak in thermal infrared bands.



Backup: Atmospheric Windows Guide Sensor Design

- ▶ Visible and NIR support reflected-solar imaging.
- ▶ Thermal IR uses transmission windows near 8–14 μm .
- ▶ Microwave sensing remains valuable under clouds and at night.
- ▶ Band selection is a physics constraint before it is an engineering preference.